

# 4.6.4

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## Question:

Find the equation of a line passing through the point  $(2, 3, 2)$  and parallel to the line  $\mathbf{r} = (-2\hat{i} + 3\hat{j}) + \lambda(2\hat{i} - 3\hat{j} + 6\hat{k})$ . Also, find the distance between these two lines.

## Solution:

### 1. Equation of the Line

The equation of the line passing through point  $\mathbf{a}_2$  parallel to vector  $\mathbf{b}$  is:

$$\mathbf{r} = \mathbf{a}_2 + \mu\mathbf{b} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} \quad (0.1)$$

### 2. Distance Between Lines using Least Squares

Let  $\Delta\mathbf{a} = \mathbf{a}_2 - \mathbf{a}_1$ . The squared distance between the lines is the minimum of the function:

$$\begin{aligned} f(\nu) &= \|\Delta\mathbf{a} - \nu\mathbf{b}\|^2 = (\Delta\mathbf{a} - \nu\mathbf{b})^T (\Delta\mathbf{a} - \nu\mathbf{b}) \\ &= \Delta\mathbf{a}^T \Delta\mathbf{a} - 2\nu(\Delta\mathbf{a}^T \mathbf{b}) + \nu^2(\mathbf{b}^T \mathbf{b}) \end{aligned} \quad (0.2)$$

To find the minimum, we set the derivative to zero:

$$\frac{df}{d\nu} = -2(\Delta\mathbf{a}^T \mathbf{b}) + 2\nu(\mathbf{b}^T \mathbf{b}) = 0 \implies \nu_{opt} = \frac{\Delta\mathbf{a}^T \mathbf{b}}{\mathbf{b}^T \mathbf{b}} \quad (0.3)$$

Substituting  $\nu_{opt}$  back gives the minimum squared distance:

$$d^2 = \frac{(\Delta\mathbf{a}^T \Delta\mathbf{a})(\mathbf{b}^T \mathbf{b}) - (\Delta\mathbf{a}^T \mathbf{b})^2}{\mathbf{b}^T \mathbf{b}} \quad (0.4)$$

By Lagrange's identity, the numerator is  $\|\Delta\mathbf{a} \times \mathbf{b}\|^2$ . This gives the distance formula:

$$d = \frac{\|\Delta\mathbf{a} \times \mathbf{b}\|}{\|\mathbf{b}\|} \quad (0.5)$$

The numerical values are:

$$\mathbf{a}_1 = \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} \quad (0.6)$$

$$\Delta\mathbf{a} = \mathbf{a}_2 - \mathbf{a}_1 = \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} \quad (0.7)$$

The cross product is:

$$\Delta \mathbf{a} \times \mathbf{b} = \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} \times \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} = \begin{pmatrix} 6 \\ -20 \\ -12 \end{pmatrix} \quad (0.8)$$

The norms are:

$$\|\Delta \mathbf{a} \times \mathbf{b}\| = \sqrt{6^2 + (-20)^2 + (-12)^2} = \sqrt{580} \quad (0.9)$$

$$\|\mathbf{b}\| = \sqrt{2^2 + (-3)^2 + 6^2} = 7 \quad (0.10)$$

The distance is:

$$d = \frac{\sqrt{580}}{7} = \frac{2\sqrt{145}}{7} \quad (0.11)$$

